Christmas demand and stockouts

How could a manager spot forecast shortfalls and review what happened when customers chose alternatives?

The starting point

While working as a Team Lead in a quick-service restaurant, I saw Christmas demand exceed expectations for some items. When stock ran short, the team promoted available substitutes.

What I built

I recreated a small fictional restaurant scenario in Excel: six menu items, two months of daily records, a weekday demand forecast and a dashboard for shortages, substitute sales and target tracking.

366
daily item records

6
fictional menu items

2
months of data

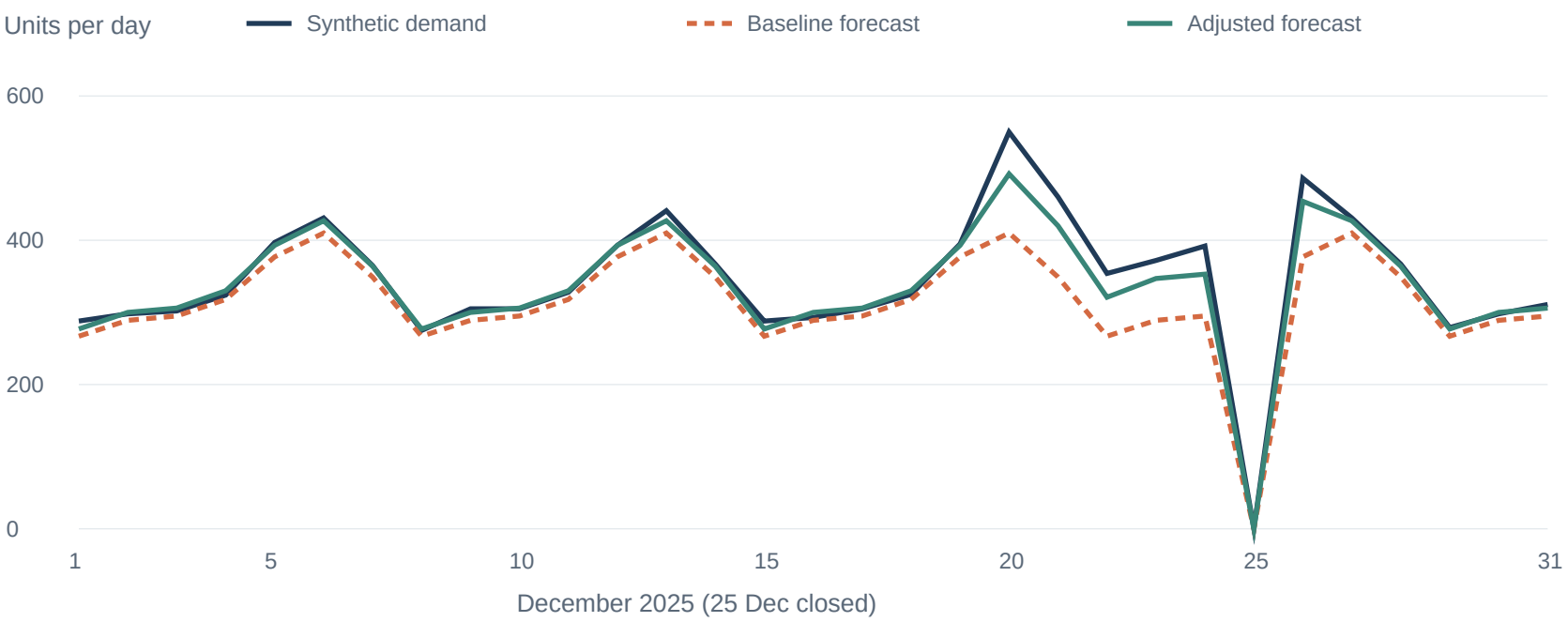
Real inspiration. Fictional numbers.
Cedar Kitchen, the dates, menu, prices and target are invented. No employer or customer records are used. I built this analysis afterwards for my portfolio; it was never used during employment.

WHAT THE SYNTHETIC SCENARIO SHOWS

Some demand was met. The target was still missed.

244 requests faced a first-choice shortage. **137** switched to substitutes; **107** remained unserved. December sales were **£69,293.50** against a fictional **£70,000** target: **£706.50 short**.

Demand and forecasts



Full-December WAPE: **9.34% baseline** and **4.61% adjusted**. WAPE measures total absolute item-day forecast error relative to demand.

The baseline uses November averages for each weekday. The adjustment adds an assumed December uplift. Because that uplift broadly matches the synthetic data pattern, lower error here is a teaching example, not proof of real-world forecasting gains.

What I would take into real analysis

Sales alone can hide unmet demand during stockouts. I would check availability records and demand estimates, then test forecasts on more history. Substitute revenue is already part of sales; assumed switching does not establish promotion effectiveness.